nature portfolio

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Xenium Analyzer instrument used for spatial transcriptomic data collection. BD FACSCanto II was used for flow cytometry data collection. EVOS M7000 was used for image acquisition. Bio-rad ChemiDoc used for western blot imaging. Spectramax M3 was used for absorbance readings.
Data analysis	FlowJo_v10.10.0, GraphPad Prism 10, ImageJ 1.54g, Xenium Explorer v3.2, Tessera (open source), Loupe v9.0. R (v4.4.1) was used with the following packages: Seurat v5.0, DESeq2 v3.2, Harmony v1.2.3, limma v3.20, eks v1.0.7

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Previously published publicly available data used in this study are available in the following repositories: systemic sclerosis single-cell RNA sequencing data

(GSE195452), idiopathic pulmonary fibrosis single-cell RNA sequencing data (GSE136831), synovial single-cell RNA sequencing data (https://doi.org/10.7303/syn52297840), micromass organoid single-cell RNA sequencing data (https://www.immport.org; SDY1599), PEAC data (E-MTAB-6141), RBPJ Chip-seq data (GSE127388; ENCSR596FEL). All spatial transcriptomic data generated as part of this study are available on Zenodo: RA synovial pre-treatment data: https://doi.org/10.5281/zenodo.15058262. RA synovial post-treatment data: https://doi.org/10.5281/zenodo.15058715. RA synovial custom fibrogenesis panel (n = 4 tissues) and AMP Olink matrix containing NPX values and clinical metadata: https://doi.org/10.5281/zenodo.15059437. Xenium profiling of 2D co-culture: https://doi.org/10.5281/zenodo.15064564. RA-ILD Lung and RA synovium Visum data, RA patient-derived organoid Xenium data, 3D co-culture Xenium Data, 396.10 Olink data: https://doi.org/10.5281/zenodo.16995699.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	The sexes of human participants has been reported in Supplementary Table 1. Sex was determined by self-reporting. Sex as a biological variable was not considered in this study.
Reporting on race, ethnicity, or other socially relevant groupings	Data about race or ethnicity was not collected or included as part of the study design or analyses.
Population characteristics	Population characteristics of the human participants is listed in Supplementary Table 1.
Recruitment	Synovium from patients with RA or OA were retrieved via biopsy, arthroplasty or synovectomy at Flinders Medical Centre, Brigham and Women's Hospital, University of Colorado Anschutz medical center and Hospital for Special Surgery. All participants were enrolled on a voluntary basis and provided written consent.
Ethics oversight	Southern Adelaide Clinical Human Research Ethics Committee (Protocol #396.10), the Mass General Brigham Human Research Committee (IRB 2019P002924), the Colorado Multiple Institutional Review Board (IRB 20-1908), and Hospital for Special Surgery (IRB 2014-233).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical methods were used to determine sample size. For spatial transcriptomic data, sample size was determined based on contemporary spatial studies. For imaging analysis, sample size (number of cells quantified) was determined in order to robustly capture the distribution of cell phenotypes observed in vitro. For in vitro experiments, sample size was determined based on preliminary experiments and previous publications.
Data exclusions	Single-cell data derived from spatial transcriptomics was filtered to remove doublets and cells with low RNA as described in methods. Prior to data integration and niche analysis, one sample was excluded due to transcript sparsity.
Replication	For spatial transcriptomic data, the observations were validated across 2 independent patient cohorts, as described in the text. The reproducibility of all in vitro data was confirmed with at least two independent experiments.
Randomization	Paired pre- and post-treatment synovial samples from the same patient and same joint were run on the same spatial transcriptomic imaging slide in order to reduce potential batch effects. Randomization n/a for other aspects of the study.
Blinding	No blinding was performed as there was no intervention.

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	*Briefly describe the study type including whether data are quantitative, qualitative, or mixed-methods (e.g. qualitative cross-sectional, quantitative experimental, mixed-methods case study).*
Research sample	*State the research sample (e.g. Harvard university undergraduates, villagers in rural India) and provide relevant demographic*

Research sample	*information (e.g. age, sex) and indicate whether the sample is representative. Provide a rationale for the study sample chosen. For studies involving existing datasets, please describe the dataset and source.*
Sampling strategy	*Describe the sampling procedure (e.g. random, snowball, stratified, convenience). Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient. For qualitative data, please indicate whether data saturation was considered, and what criteria were used to decide that no further sampling was needed.*
Data collection	*Provide details about the data collection procedure, including the instruments or devices used to record the data (e.g. pen and paper, computer, eye tracker, video or audio equipment) whether anyone was present besides the participant(s) and the researcher, and whether the researcher was blind to experimental condition and/or the study hypothesis during data collection.*
Timing	*Indicate the start and stop dates of data collection. If there is a gap between collection periods, state the dates for each sample cohort.*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, provide the exact number of exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Non-participation	*State how many participants dropped out/declined participation and the reason(s) given OR provide response rate OR state that no participants dropped out/declined participation.*
Randomization	*If participants were not allocated into experimental groups, state so OR describe how participants were allocated to groups, and if allocation was not random, describe how covariates were controlled.*

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	*Briefly describe the study. For quantitative data include treatment factors and interactions, design structure (e.g. factorial, nested, hierarchical), nature and number of experimental units and replicates.*
Research sample	*Describe the research sample (e.g. a group of tagged Passer domesticus, all Stenocereus thurberi within Organ Pipe Cactus National Monument), and provide a rationale for the sample choice. When relevant, describe the organism taxa, source, sex, age range and any manipulations. State what population the sample is meant to represent when applicable. For studies involving existing datasets, describe the data and its source.*
Sampling strategy	*Note the sampling procedure. Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.*
Data collection	*Describe the data collection procedure, including who recorded the data and how.*
Timing and spatial scale	*Indicate the start and stop dates of data collection, noting the frequency and periodicity of sampling and providing a rationale for these choices. If there is a gap between collection periods, state the dates for each sample cohort. Specify the spatial scale from which the data are taken*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Reproducibility	*Describe the measures taken to verify the reproducibility of experimental findings. For each experiment, note whether any attempts to repeat the experiment failed OR state that all attempts to repeat the experiment were successful.*
Randomization	*Describe how samples/organisms/participants were allocated into groups. If allocation was not random, describe how covariates were controlled. If this is not relevant to your study, explain why.*
Blinding	*Describe the extent of blinding used during data acquisition and analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.*

Did the study involve field work? ☐ Yes ☐ No

Field work, collection and transport

Field conditions	*Describe the study conditions for field work, providing relevant parameters (e.g. temperature, rainfall).*
Location	*State the location of the sampling or experiment, providing relevant parameters (e.g. latitude and longitude, elevation, water depth).*
Access & import/export	*Describe the efforts you have made to access habitats and to collect and import/export your samples in a responsible manner and in compliance with local, national and international laws, noting any permits that were obtained (give the name of the issuing authority, the date of issue, and any identifying information).*

Disturbance | *Describe any disturbance caused by the study and how it was minimized.*

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Immunofluorescence : Phospho-SMAD3 (Ser423, Ser425) (16H5L12), Thermofisher Cat. 702292; TGF beta-1 (TB21), Thermofisher Cat. MA5-18023; TGF beta-2 (220ct16.4.3.1), Thermofisher Cat. MA5-37505; TGF beta-3, Thermofisher Cat. PA1-38736; PE anti-human Notch 3 (MHN3-21), Biolegend Cat. 345405; FITC anti-human CD45 (2D1), Biolegend Cat. 368507; Collagen I (COL-1), Thermofisher Cat. MA1-26771; VWF (F8/86), Thermofisher Cat. MA5-14029; Goat anti-Mouse IgG Alexa Fluor 660, Thermofisher Cat. A-21055; Donkey anti-Rabbit IgG Secondary Antibody Alexa Fluor 555, Thermofisher Cat. A-31572.

Flow Cytometry: APC anti-human TGF-β Receptor II, Biolgend Cat. 399705; TGFBR3, Thermofisher Cat. 20000-1-AP; FITC anti-human CD31 (WM59), Biolegend 303103. anti-Rabbit IgG Secondary Antibody Alexa Fluor 555, Thermofisher Cat. A-31572. BV421-CD31, Biolegend Cat. 303123, FITC-CD45 Biolegend Cat. 368507, PE-PDPN Biolegend Cat. 337003

Western Blot: TGF Beta 1, Proteintech Cat. 21898-1-AP; TGF Beta 2, Proteintech Cat. 19999-1-AP; TGF Beta 3, Proteintech Cat. 18942-1-AP; Human TGF-beta RI/ALK-5, R&D Systems Cat. AF3025; TGF beta Receptor 2, Bioss Cat. BS-0117R; TGF-β Receptor III, Cell Signaling Technology 2591S; GAPDH, Thermofisher Cat. MA5-15738

Validation

All antibodies used are commercially available and have been validated by the manufacturer. Validation data is available on the manufacturer's website.

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)

Primary synovial fibroblast cell lines isolated from OA or RA patients were used. HUVECs were attained from Thermofisher (https://www.thermofisher.com/order/catalog/product/C0035C)

Authentication

None of the cell lines were authenticated

Mycoplasma contamination

Cells lines were not tested for mycoplasma.

Commonly misidentified lines
(See ICLAC register)

Not applicable

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	Fibroblasts or co-culture of fibroblasts and endothelial cells were trypsinized and directly stained with an equal volume of primary antibody solution (Cell staining buffer, Biolegend) containing eFluor 780 Fixable Viability Dye (Thermofisher, 1:1000), TGFBR2-APC (Biolegend, 1:100), TGFBR3 (Proteintech, 20000-1-AP, 1:100) or CD31-FITC (Biolegend, 1:100) for 30 minutes at 4C. Cells were washed with cell staining buffer (Biolegend), fixed in 4% PFA for 10 minutes at RT, washed, and stained with AF555 anti-rabbit (Thermofisher) for 30 minutes at 4°C for detection of TGFBR3. Cells were washed and resuspended in cell staining buffer.
Instrument	BD FACSCanto II
Software	BD FACSDiva and Flowjo v10.10.0
Cell population abundance	Cells were not sorted prior to flow cytometry or experiments
Gating strategy	Fibroblasts gating: FSC-A/SSC-A plots were used to determine cell population gates. FSC-A/FSC-H plots were then used to determine singlet gates. Cells that were negative for the live/dead stain and negative for CD31 were selected for analysis.

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.